

SOLAR-TWIN · SLICE 0 → SLICE 3

What the twin has actually measured

An autonomous solar-farm inspection loop running headless in Isaac Sim 6.0.1 on a DGX Spark, with a live Cosmos Reason VLM as the detector. Every number below is read out of a committed run record in `runs/`; every frame is re-rendered from the current USD stage for this readout.

18 recorded runs

10 live-VLM missions

74 tests pass, 2 skipped

read 27 Jul 2026

[theme](#)

KPI-01 · DETECTION

1.00

10/10 panels correctly stated, both seeded soiled panels named. Run ...T172011.

KPI-03 · FALSE FAULTS

0.00

No healthy panel faulted, on three shadow scenarios. Gate is ≤ 0.05 — but read the coverage caveat below.

KEEP-OUT CLEARANCE

9.42 m

Min turbine clearance on the real plan; 0 waypoints clamped, 2 turbines.

MISSION WALL TIME

89 s

10 panels, screen + confirm passes, live VLM. 1.5–3.3 s with the ground-truth stub.

THE LOOP

What one run does

Six stages, no GUI step anywhere — a run is fully described by a config pair, a seed and a build.

01

Author the world

`farm_builder.py` writes a USD farm from `farm.yaml`: terrain grade, tilted PV modules, seeded faults, turbines.

02

Fly the row

Kinematic ground bot + two drones, waypoints clamped by the turbine keep-out volume before any motion.

03

Screen

Wide pass 2.5 m over the panel top. Camera RGB goes to the detector as a PNG data URL.

04

Escalate + confirm

Suspect panels get a second pass at 0.8 m; the FSM asks for a named fault, not a yes/no.

05

Write back

Verdicts land on the USD prims themselves (`pv:state`, `pv:inspection_log`) — the stage is the record.

06

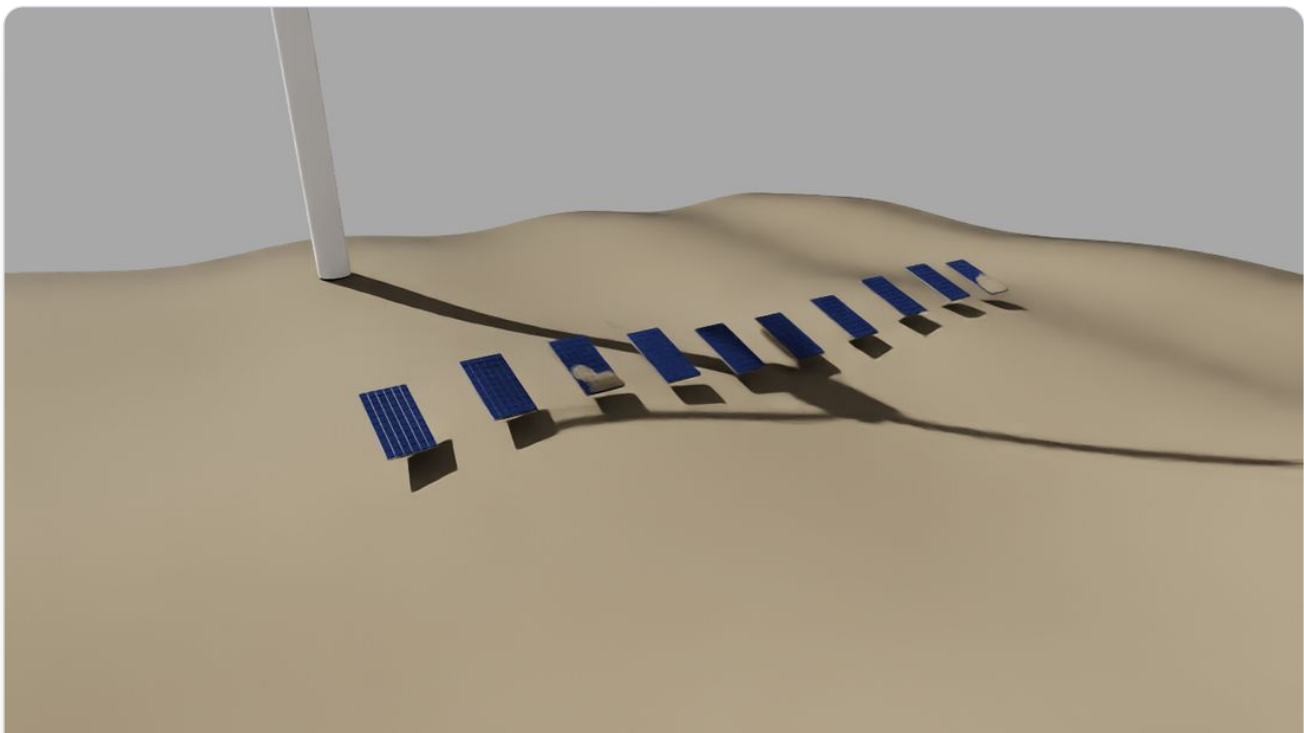
Emit the record

Injected-vs-detected, KPIs, keep-out audit and the model's own reasoning to `runs/<ts>/results.json`.

THE WORLD

Ten panels on graded ground, under turbines

Re-rendered from `assets/farm.usd` for this readout. The two tan panels are the seeded soiled pair; the dark sweep across the row is a turbine blade shadow — the same moving occluder the false-fault harness is built around.



Nominal farm — seed 20260721

Panels mount on a deterministic height field rather than flat ground, and drone standoffs are measured from each panel's own top face, so framing survives the grade.

KPI - 01

Detection went 0.80 → 1.00, and none of the four fixes were in the model

Seven live-VLM runs over the same seeded farm. Each bar is ten panels, split by what the pipeline concluded about them. The story the bars tell is a two-stage failure: first the faults were invisible, then they were visible but misnamed.

Correct — matches ground truth

Escalated, wrong fault name

Fault missed entirely

right column = detection rate

20260721T190558
21 Jul

8

2

0.80

First live Cosmos Reason run. Panels are flat colour swatches — the model replies “not a photograph of a solar panel.”

20260724T162536
24 Jul

8

2

0.80

Render fix: real PV cell grid + aluminium frame, directional sun, wider FOV, panel-top-relative standoffs. Now recognised as a module.

20260724T164326
24 Jul

8

2

0.80

Keep-out debug spheres re-authored as `purpose="guide"` . They had been shadowing the whole farm — frame brightness 46 → 126.

20260724T170822
24 Jul

8

2

0.80

Soiling rebuilt as a translucent film that crosses cell borders. Both soiled panels finally escalate — but come back labelled *hotspot*.

20260724T171133
24 Jul

8

2

0.80

Dust alpha and bake resolution tuned (sub-grid ≈ 8× the cell grid; RTX renders preview-surface opacity as a hard cutout).

20260724T171600
24 Jul

8

2

0.80

Further appearance tuning. Escalation holds at 2/2; the label is still wrong, so

detection rate will not move.

20260724T172011
24 Jul

10

1.00

The fix was the **prompt**, not the world: the diagnose step passed bare enum names with no definitions, so any localised anomaly mapped to the model's hotspot prior.

1 Flat colour swatches, not photographs

Panels
were
single-
colour
boxes.
The
model
said
so
plainly
—
*"not
a
photograph
of a
solar
panel"*.
Fixed
with
a real
cell
grid,
an
aluminium
frame,
directional
sun
and a
wider
camera
FOV.

2 A debug aid was corrupting the sensor path

The

